EEE141L: ELECTRICAL CIRCUITS LAB

LAB: 06

EXPERIMENT NAME: VERIFICATION OF SUPERPOSITION THEOREM

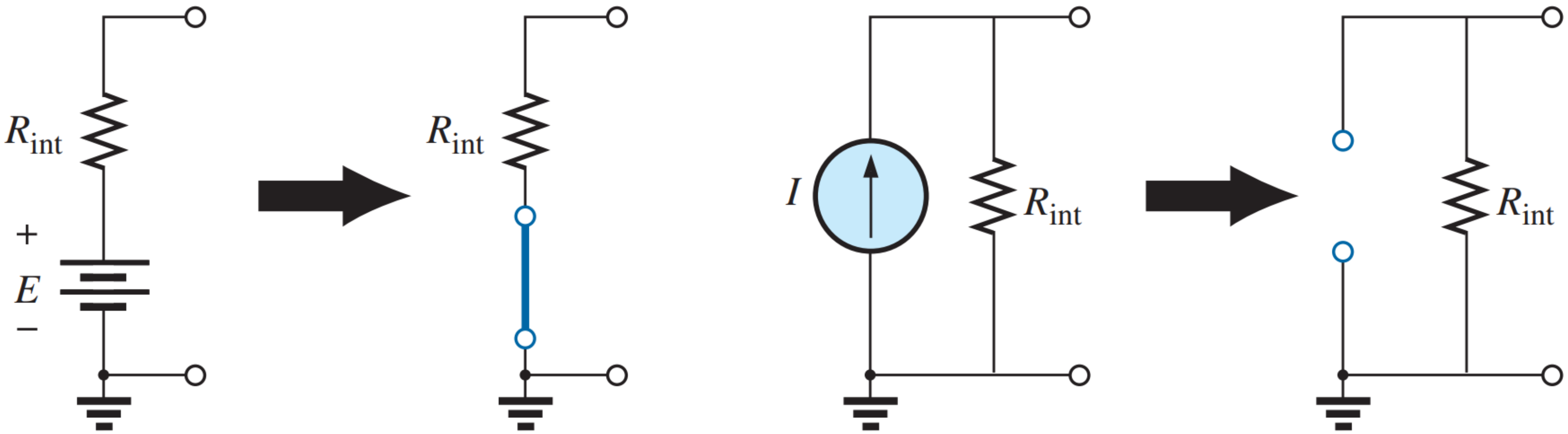
LAB INSTRUCTOR: SUDIPTO SARKER SUPTO

SUPERPOSITION THEOREM

- If a circuit has two or more independent sources, one way to determine the value of a specific variable (voltage or current) is to use nodal or mesh analysis.
- Another way is to determine the contribution of each independent source to the variable and then add them up. The latter approach is known as the superposition.
- **The superposition theorem states that the current through, or voltage across, any element of a network is equal to the algebraic sum of the currents or voltages produced independently by each source.**
- The principle of superposition helps us to analyze a linear circuit with more than one independent source by calculating the contribution of each independent source separately.

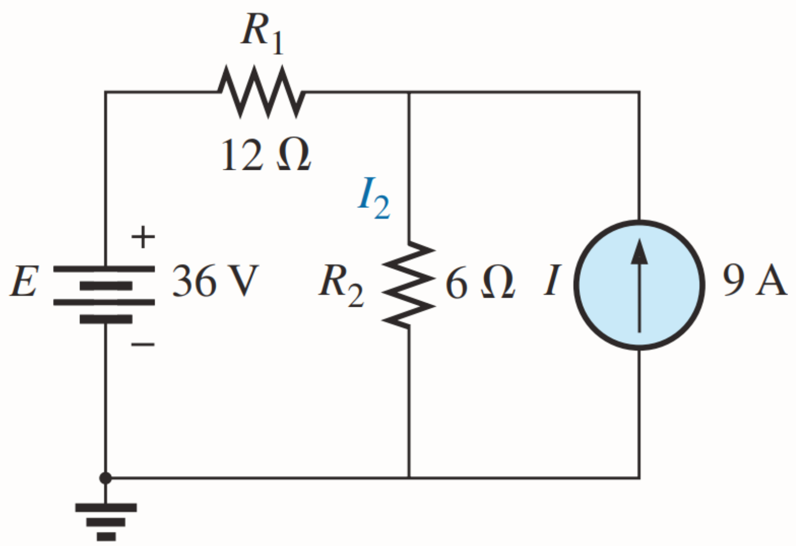
SUPERPOSITION THEOREM
CONTINUED...

- If we are to consider the effects of each source, the other sources obviously must be removed.
- Setting a voltage source to zero volts is like replacing it with a short circuit of zero ohms; setting a current source to zero amperes is like replacing it with an open circuit of infinite ohms.

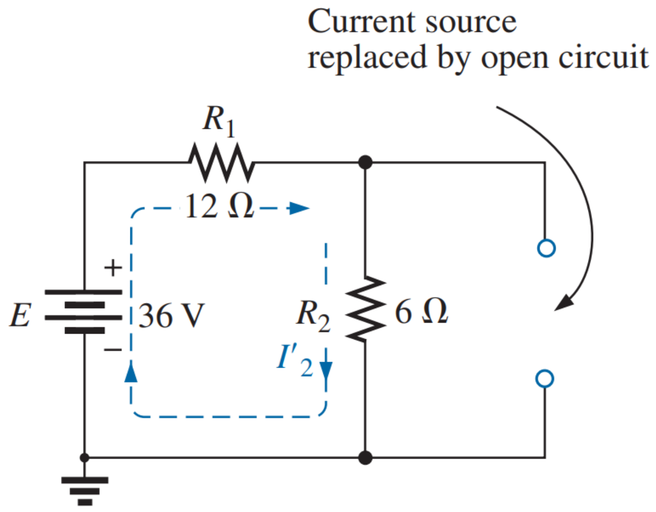


SOLVING USING SUPERPOSITION THEOREM

Using the superposition theorem, determine the current through resistor R_2 for the network.



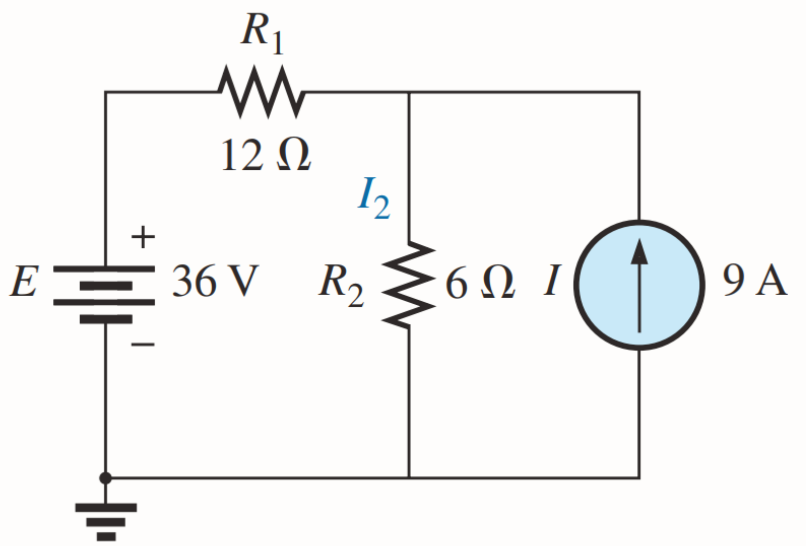
In order to determine the effect of the **36 V** voltage source, the current source must be replaced by an open-circuit equivalent.



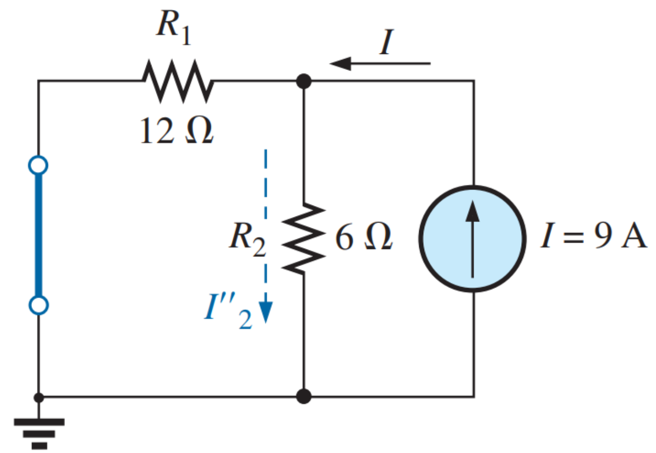
$$I_2' = \frac{E}{R_T} = \frac{E}{R_1 + R_2} = \frac{36\ \text{V}}{12\ \Omega + 6\ \Omega} = \frac{36\ \text{V}}{18\ \Omega} = 2\ \text{A}$$

SOLVING USING SUPERPOSITION THEOREM
CONTINUED...

Using the superposition theorem, determine the current through resistor R_2 for the network.



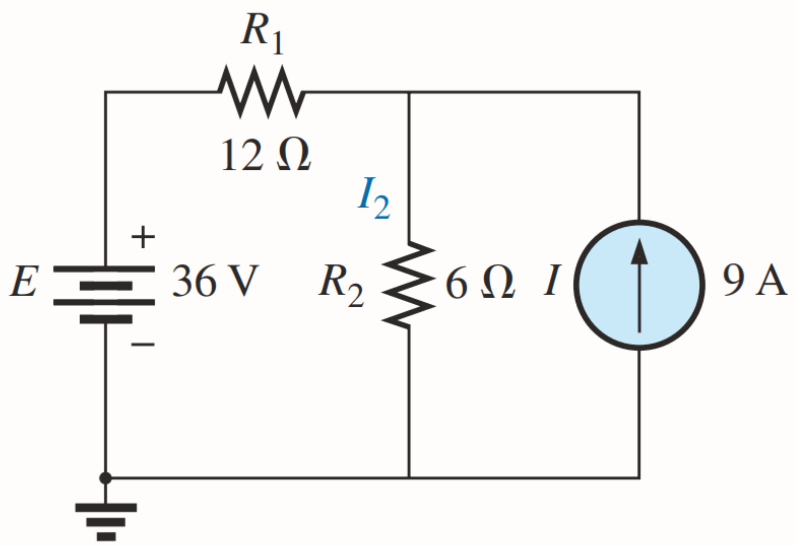
In order to determine the effect of the **9 A** current source, the voltage source must be replaced by an short-circuit equivalent.



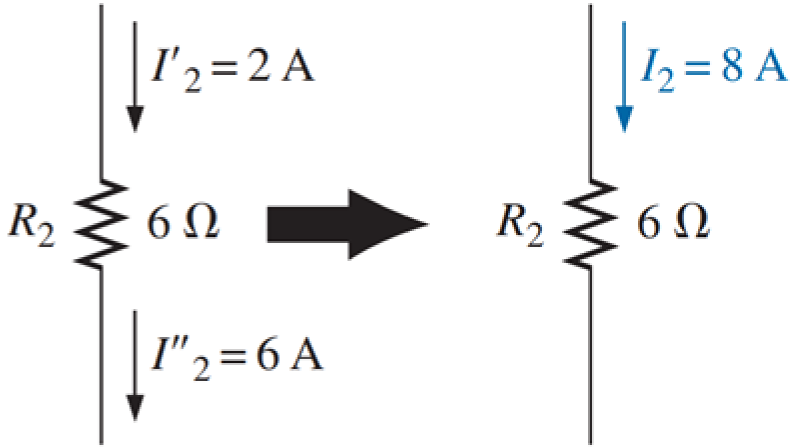
$$I_2'' = \frac{R_1(I)}{R_1 + R_2} = \frac{(12\ \Omega)(9\ \text{A})}{12\ \Omega + 6\ \Omega} = 6\ \text{A}$$

SOLVING USING SUPERPOSITION THEOREM
CONTINUED...

Using the superposition theorem, determine the current through resistor R_2 for the network.



Since the contribution to current I_2 has the same direction for each source, the total solution for current I_2 is the sum of the currents established by the two sources.



$$I_2 = I'_2 + I''_2 = 2\text{ A} + 6\text{ A} = \mathbf{8\text{ A}}$$

THE END...

Prepared by: Sudipto Sarker Supto